Transcriptomic signatures of Extracellular Vesicles from Alzheimer's Disease iPSC-derived Neurons

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INTRODUCTION

- Alzheimer's disease (AD) profoundly impacts over 55 million individuals globally, with cases projected to triple by 2050, posing an escalating public health challenge. The underlying mechanisms of AD remain incompletely understood, highlighting the urgent need for reliable biomarkers for early detection of its pathological features. Extracellular Vesicles (EVs) are lipid bilayer membrane-enclosed nanoparticles of cellular origin, instrumental in facilitating various intercellular interactions through the transport and transfer of molecules between neighboring cells. EVs have come to light as important factors containing cargo that may alter/contribute to AD pathogenesis.
- The aim of this study was to compare the cargo composition of EVs isolated from neurons of people with AD to controls. Induced Pluripotent Stem Cells (iPSCs) were derived from patients with AD and cognitively unimpaired (CU) individuals , followed by differentiation into excitatory glutamatergic neurons. Subsequently, secreted EVs were isolated from these mature neurons, followed by the execution of bulk RNA sequencing.

MATERIALS AND METHODS

- The hiPSCs were differentiated into glutamatergic excitatory neurons by using NgN2 induction protocol and culture supernatant was collected from mature neurons.
- Size exclusion chromatography (SEC) using qEV columns was performed for EVs isolation from culture's supernatant
- EV yield and size distribution were determined by Nanoparticle Tracking Analysis, their morphology was evaluated by transmission electron microscopy. Further characterization was performed by Heatmap of Single Particle Interferometric Reflectance Imaging Sensing (SP-IRIS) platform for surface markers (CD9, CD63, CD81).
- RNA was extracted from iPSC-derived neuronal extracellular vesicles (EVs) and were sequenced using the NovaSeq 6000 system

RESULTS

The reprogramming of iPSCs was performed from a total of 14 samples of PBMCs originating from either healthy individuals (n=6) or AD patients (n=8) and differentiated into excitatory glutamatergic neurons. Transcriptomic analysis revealed the presence of various RNA biotypes within the EVs, notably highlighting a significant distinction in small nuclear RNAs (RNU) between EVs derived from AD versus CU. At adjusted p<0.05, 57 genes were differentially expressed between the EVs from AD and CU neurons. Visualization of the transcriptomic data showed significantly upregulated genes within EVs from AD neurons, associated with pathways related to neuronal maturation, epigenetic modifications, and mitochondrial functions.

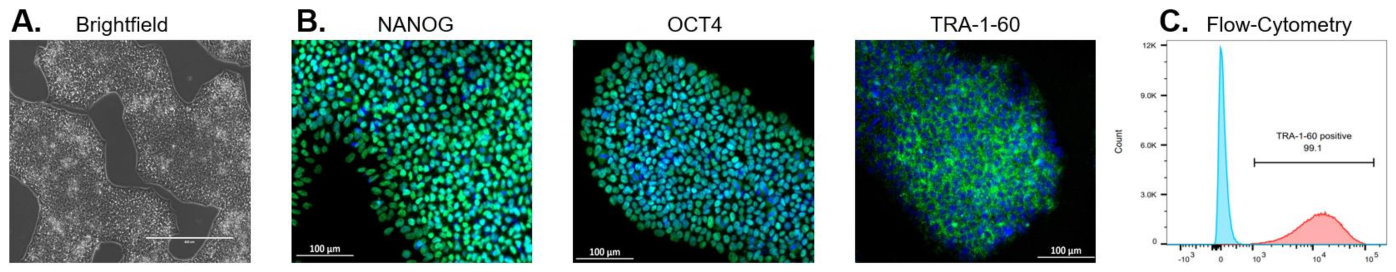

Figure 1: **Generation of iPSCs. (A)** The morphology of iPSCs is shown in a representative brightfield microscopy image. **(B)** Fluorescent microscopy for pluripotency markers TRA-1-60, OCT4, NANOG. **(C)** Flow cytometry was additionally used to show the purity of TRA-1-60 positive cells in all cultures.

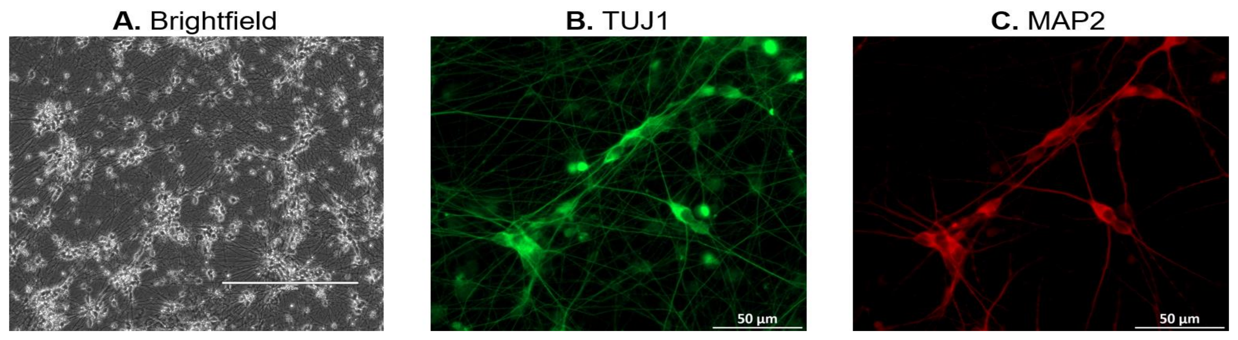

Figure 2: (A) Representative brightfield and fluorescent images for neuronal specific markers **(B)** TUJ1 & **(C)** MAP2 on differentiated neurons.

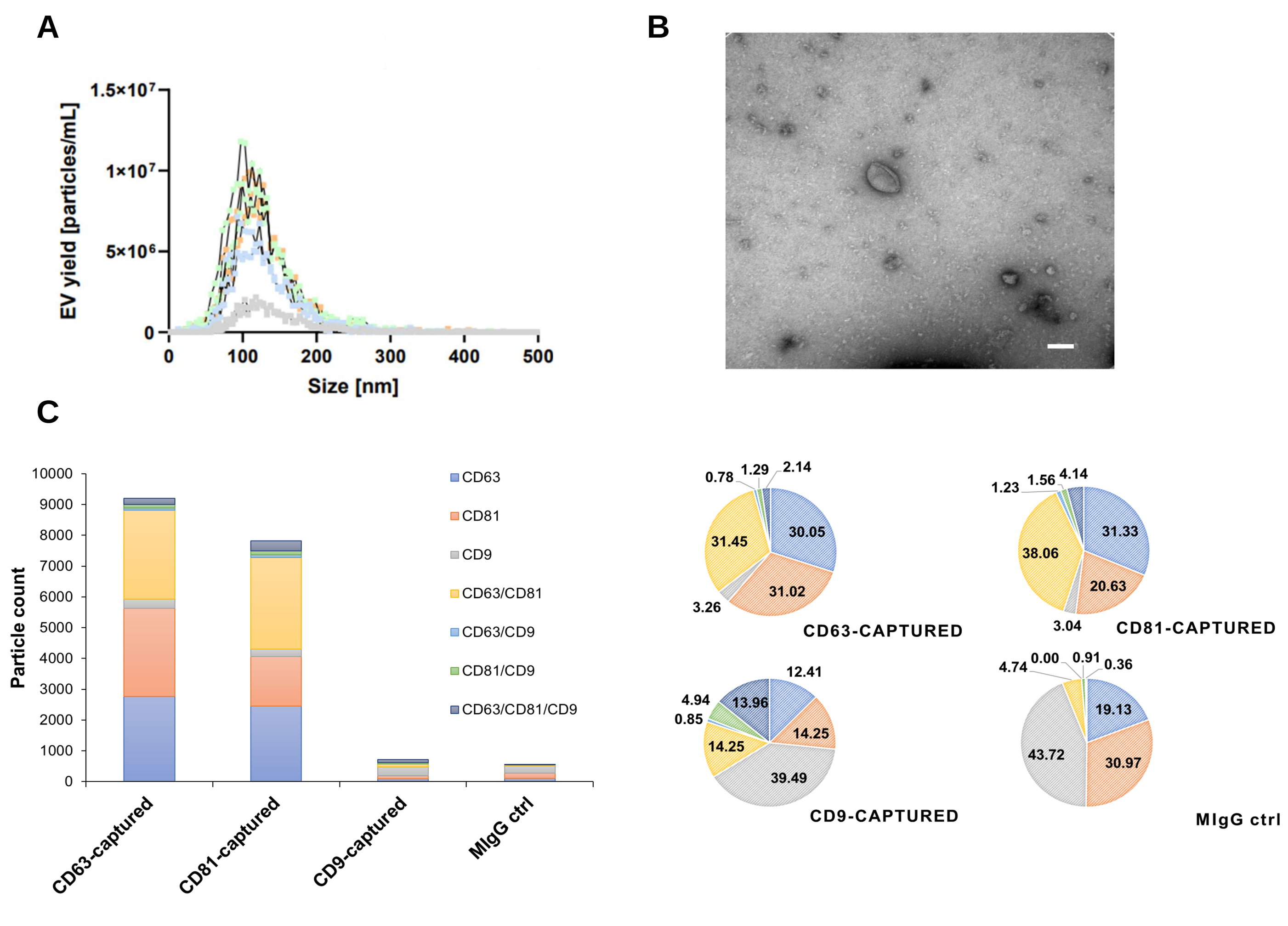

Figure 3: **(A)** Representative size distribution profiles for EVs at different harvesting time points as measured by NTA. **(B)** EVs were further evaluated by TEM & **(C)** SP-IRIS.

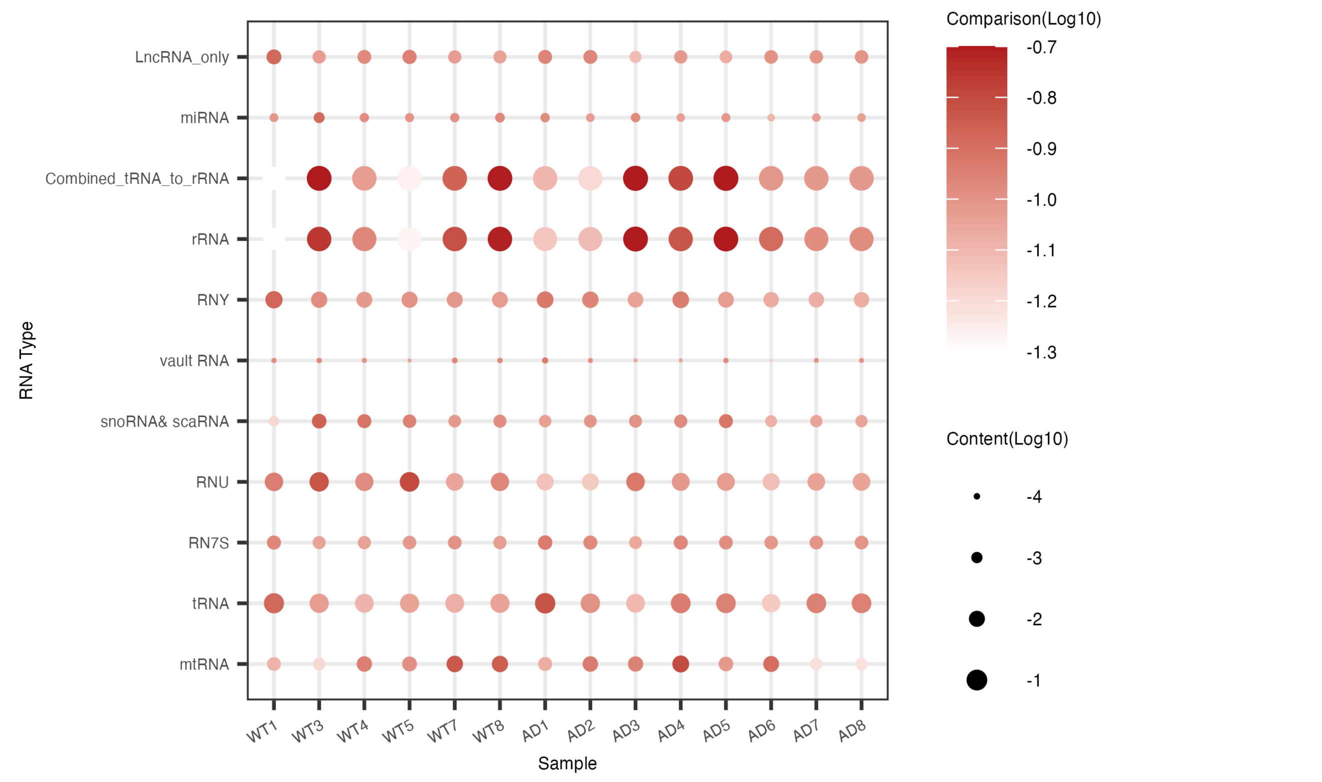

Figure 4: Proportion of RNA present in each EV samples. Content represents the percentage of RNA type in each of the samples. The darker color shows that higher percentage of RNA is observed in the EV cargo of the samples.

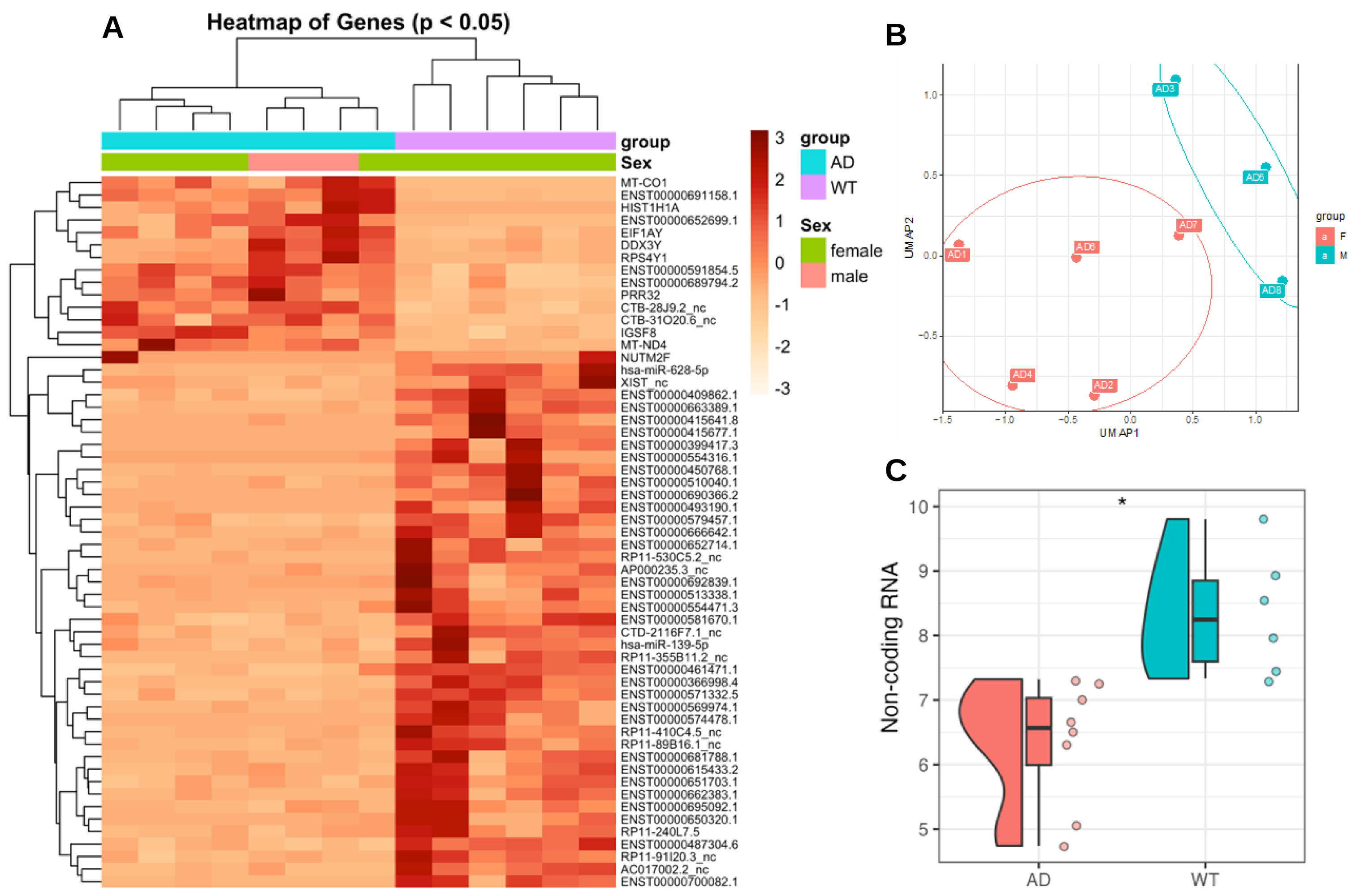

Figure 5: **(A)** Heatmap of all samples based on differentially expressed genes between AD EV and WT EV groups with p < 0.05. **(B)** UMAP clustering analysis for AD EV samples. **(C)** Differences in a non-coding RNA between AD EV and WT EV groups.

CONCLUSION

Extracellular Vesicles were isolated from glutamatergic excitatory neurons generated from hiPSCs derived from both healthy and patients diagnosed with AD. Analysis of the bulk RNA sequencing data revealed notable variations in gene expression between the two groups. Further exploration indicated the formation of distinct patient clusters based on their gene expression patterns. Our findings further substantiate the evidence that EV cargo from AD neurons exhibit alterations that may contribute to elucidate AD pathology and discovery of disease biomarkers.

ACKNOWLEDGEMENT

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